

Calibrated Uncertainty for Active Room-Layout Estimation

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Abstract—A robot that estimates the layout of a room usually reports geometry and little else. Downstream consumers—planners, scene graphs, other agents—must then decide how far to trust it, with nothing to go on. We present an online room-layout estimator whose reported uncertainty is calibrated against ground truth, and which uses that uncertainty as a decision variable: it gates which corners are published, and it is what the robot acts to reduce. Parameters, structure (how many walls there are) and viewpoint selection are all obtained by minimising one variational free energy, so the same quantity in nats prices the data fit, the model complexity and the next observation. On 594 annotated floor plans the estimator reaches 0.983 mean IoU, and—the claim of the paper—a corner it declares publishable has a median true error of 2.0 cm and lies within its declared 6 cm in 86 % of cases, against 3.6 cm and 64 % for corners it withholds. Holding the system to that standard proved diagnostic: it exposed three distinct precision defects in three different modules, all instances of uncertainty that averages away past an error that does not, one of them in a module the instrument was not measuring. We show that no scalar precision can be correct for this problem—the irreducible error varies by $5\times$ between environments and $2\times$ between moving and standing still—and argue that this is why precision must be inferred rather than tuned.

I. INTRODUCTION

Room-layout estimation is mature enough that accuracy alone no longer separates methods. What separates them for a robot is whether the estimate can be *acted on*: whether a planner can tell a wall that is known to a centimetre from one that is a guess, whether a scene graph should commit a corner to shared memory, whether the robot should look again before deciding. All of these are questions about uncertainty, and almost no layout system answers them with a number that has been checked.

This is not an oversight of presentation. An uncertainty that is never validated tends to be wrong, and wrong in a way that is invisible: an estimator that overstates its error is merely timid, while one that understates it is confidently incorrect, and neither shows up in a mean IoU. We found both failure modes in our own system, and we found them only because something downstream was made to depend on the number being right.

Our estimator maintains a room as a cyclic polygon of wall landmarks in Hesse normal form, inferred jointly with the robot pose over a sliding window. Three decisions that are usually taken by three separate mechanisms are here taken by one. *What the walls are* is variational inference over their parameters. *How many walls there are* is Bayesian model comparison: a structural change is adopted when it pays for

its own description length, priced in nats on the same scale as the data fit, rather than by a merge threshold. *Where to look next* is the epistemic term of the expected free energy, so the robot moves to reduce the uncertainty that is currently blocking it from committing to a layout. Because all three are denominated in the same currency, every precision in the model is load-bearing: get one wrong and the comparisons it feeds are priced in a corrupt unit. That is the property we exploit.

Contributions.

- 1) An online layout estimator whose per-corner and per-edge uncertainty is *validated against ground truth* on 594 annotated floor plans, and is *actionable*: it gates publication and drives exploration (§V).
- 2) A propagation from returns to corner covariance that separates the error that averages down with more observation from the error that does not, and a measurement of both terms on real sensor data (§IV).
- 3) Evidence that a scalar precision cannot be calibrated for this problem: the irreducible component varies by $5\times$ across environments and $2\times$ between motion states, which we argue is a general property of map-relative estimation rather than a quirk of our system (§V-D).
- 4) Three precision defects found by calibration, including one localised in a module the statistic was not measuring—offered as evidence that a free-energy formulation earns its keep diagnostically, not just descriptively.

II. RELATED WORK

Learned layout estimation. Panorama-based methods recover room shape from a single view and are strong under Manhattan or Atlanta priors. They produce a layout per image, not a belief a robot can accumulate, and report no per-corner uncertainty that a consumer could gate on.

Floor-plan reconstruction. Methods that reconstruct floor plans from accumulated point clouds or density maps optimise a global objective over a whole scan and are evaluated by corner and room F1. They are typically offline, and the structural decision—how many corners—is taken by a search with hand-set costs rather than a likelihood.

Structured SLAM. Situational and hierarchical scene-graph SLAM systems introduce planes, rooms and floors as explicit entities and demonstrate the benefit of structure for drift. Their covariances are those of the underlying least-squares problem and, to our knowledge, are not validated

against layout ground truth; the model error between a mapped plane and the real surface is not represented.

Uncertainty and calibration. Conformal prediction has recently been applied to LiDAR-based mapping to produce distribution-free guarantees. The guarantee is on a coverage set rather than on a covariance a estimator can consume, and the underlying precision is not itself audited. Our position is complementary: we keep a parametric covariance, because that is what a joint estimator and a consumer both need, and we check it.

Active perception. Next-best-view and active SLAM select viewpoints by information gain. Our selection rule is the epistemic term of expected free energy, which is the same quantity when the model is Gaussian; the contribution here is not the rule but that the precision it consumes is verified, so the gain it predicts is a gain that exists.

III. GENERATIVE MODEL

A room is a cyclic sequence of wall landmarks. Wall w is a line in Hesse normal form, $\mathbf{n}(\phi_w) \cdot \mathbf{p} = d_w$, with belief $\mathcal{N}([\phi_w, d_w]^\top, \Lambda_w^{-1})$. The robot pose over a sliding window of K frames is $x_{1:K}$. Given returns z , perception minimises the variational free energy

$$F = \underbrace{-\log p(z | x, \Phi)}_{\text{accuracy}} + \underbrace{D_{\text{KL}}(q(x, \Phi) \| p(x, \Phi))}_{\text{complexity}}, \quad (1)$$

which under a Gaussian q reduces to the sparse nonlinear least-squares problem we solve by Gauss–Newton over the window.

Structure as model comparison. The number of walls is not fixed. A candidate structural change—birth, death, or the splice of a new edge into the cycle—is a different model m' , and is adopted when $F(m') < F(m)$. The complexity term charges each additional edge a fixed description length: with a room of radius R discretised at resolution c , an edge costs $\ln 4 + \ln(2R/c)$ nats to name. There is no merge threshold and no minimum wall length; a thin partition survives exactly when the returns that support it pay for the nats it costs.

Action. A viewpoint v is scored by the expected information gain $\mathbb{E}_o[H(x, \Phi) - H(x, \Phi | o)]$, predicted through the model’s own forward sensor: beams are cast against the current belief, and each wall’s expected Fisher contribution $J_w(v)$ gives a Gaussian gain $\frac{1}{2} \ln(\det(\Lambda_w + J_w) / \det \Lambda_w)$. Because publication is gated on corner σ (§IV), the robot acts, in effect, to make its own map publishable.

IV. WHAT A LAYOUT ESTIMATE IS WORTH

A corner is the intersection of two walls, so its covariance follows from theirs. With $M = [\mathbf{n}_a^\top; \mathbf{n}_b^\top]$ and the intersection $\mathbf{p}$,

$$\Sigma_{\mathbf{p}} = J_a \Lambda_a^{-1} J_a^\top + J_b \Lambda_b^{-1} J_b^\top + \sigma_m^2 I, \quad (2)$$

where J ’s ϕ column carries the lever arm $\mathbf{t} \cdot \mathbf{p}$ and M^{-1} contributes $1/\sin \theta$ between the walls, so a shallow corner is correctly reported as uncertain along its bisector without any angle test.

The term that matters for this paper is the last one. The first two shrink as $1/N$ with the returns behind each wall: observe

TABLE I
LAYOUT ACCURACY, 594 ANNOTATED FLOOR PLANS, ONE RUN.

IoU, mean	0.9827
IoU, median	0.9870
IoU, 10th percentile	0.9690
IoU ≥ 0.95	95.2 %
IoU ≥ 0.90	98.8 %
Hausdorff, median (m)	0.0705
Coverage of truth interior, mean	0.9996
Failures (no closed polygon)	8 /594

TABLE II
CALIBRATION OF THE PUBLISHED CORNER σ AGAINST GROUND TRUTH.
4344 MATCHED CORNERS. σ IS THE LARGEST-AXIS STANDARD
DEVIATION, SO THIS TESTS A BOUND.

	$\leq 1\sigma$	$\leq 2\sigma$	$\leq 3\sigma$	publishable	claim holds
no σ_m	46.7%	70.5%	81.2%	89.0%	85.8%
$\sigma_m = 0.02$ m	65.3%	85.0%	91.2%	89.0%	86.1%

longer and they go to zero. **The true error does not follow them down.** What remains when the statistics are exhausted is systematic—the offset between the mapped wall and the real surface, identical on every look from a given viewpoint—and it is what σ_m represents. Omitting it produces an estimator that grows arbitrarily confident about a corner whose error is bounded below; double-counting it produces one that is timid. We measure it rather than tune it, and §V-D shows why it cannot be a constant.

V. EXPERIMENTS

A. Geometric generality: 594 annotated floor plans

We evaluate on 594 room annotations from a large real-building layout corpus, spanning 6 to 18 ground-truth corners. Observations are simulated from the annotated polygon; this tests the estimator against the *shape distribution of real buildings* rather than against a generator’s idea of a room, and §V-C validates the sensor model used to do it.

B. Is the reported uncertainty honest?

For every room we take the final published polygon and match its corners one-to-one to the truth corners by Hungarian assignment. Surplus corners on either side remain unmatched and are counted as *structure* errors—never as position error. This matters: nearest-vertex matching instead yields a mean error of 0.43 m against a median of 0.023 m, because a spurious corner has no truth counterpart and its “error” is meaningless.

Table II shows the effect of the systematic term of (2). Without it, the true error is flat at about 2 cm across every band of declared σ : a corner claiming 1 cm is no more accurate than one claiming 5 cm, and the declared uncertainty ranks corners without predicting their error. The measured floor is 0.019 m, against a simulated sensor of 0.020 m per return. Adding it as a variance term—not a clamp—calibrates the bound at almost no cost in yield.

TABLE III
THE TWO COMPONENTS, MEASURED ON REAL DEPTH AGAINST
ANNOTATED WALLS.

	median (m)	p90
scatter about the fitted wall (averages down)	0.0149	0.0382
fitted wall vs. annotated wall (does not)	0.0181	0.0572

The actionable claim. Publication is gated at $\sigma \leq 6$ cm. A corner the system declares publishable has a median true error of 2.0 cm and lies within the 6 cm it claims in 86.1 % of cases; a corner it withholds has a median error of 3.6 cm and satisfies the same bound only 63.7 % of the time. The gate discriminates, and the number it gates on now means something.

C. The real-sensor split

The decomposition in (2) claims that error separates into a part that averages away and a part that does not. We measure both on real data: real building-scale depth against annotated walls, fitting each wall from the returns that land on it (588 segments, 120 rooms).

Two consequences. First, the simulated sensor used in §V is honest, indeed slightly conservative: real depth scatters at 1.5 cm against the 2.0 cm simulated, so the calibration above rests on a noise model that does not flatter it. Second, the systematic component is a property of *the map*, not of the sensor—which sets up the next result.

D. Why no scalar precision can be correct

On the robot we log, for every corner candidate at the association gate, the innovation and each term of its covariance: 126 008 candidates over a 200 m tour. Normalised innovation squared (NIS, divided by its two degrees of freedom) equals 1 when a covariance matches the error it claims.

Recovering inter-frame motion from the predicted corners themselves reveals that NIS depends on what the robot is *doing*, not where it is: at rest, $NIS = 3.05$ with the gate refusing 32 % of candidates; while turning, $NIS = 1.50$. The detection covariance is flat across both ($\sigma_{\text{det}} = 0.043$ m), so the defect is upstream. Decomposing the innovation for a fixed corner across stationary frames separates it cleanly: the part that *repeats* is 0.0952 m and the part that *scatters* is 0.0113 m, a ratio of 8.1, with the bias exceeding the scatter in 97 % of cases. Averaging 1.1 cm of genuine scatter earns exactly the 1.5 cm pose uncertainty the estimator reports; the remaining 9.5 cm is a map offset represented nowhere in the model.

Combined with Table III, the systematic term measures 1.8 cm against a carefully annotated building and 9.5 cm against a hand-traced layout—a factor of 5 for the same sensor and the same estimator—and varies by a factor of 2 between standing still and moving, because in motion the per-wall offsets partially cancel in the pose solution. A constant tuned in one environment is wrong in the next; a constant tuned while the robot moves is wrong when it stops. This is the

argument for representing the offset as an inferred state with a prior rather than a compiled-in number.

E. Calibration as a diagnostic

Requiring precisions to mean what they say found three defects of one kind—uncertainty that averages down past an error that does not. In the layout channel the systematic term was absent (Table II). In the localisation channel it was present twice under two names, making the channel $2.1\times$ too timid; removing the duplicate and measuring the survivor moved NIS from 0.47 to 1.03. In the pose estimator it is absent again, and was found by an instrument pointed at the layout. We regard the third as the strongest evidence that the formulation does work rather than description: nothing in a pipeline of separately-tuned heuristics depends on a precision being right, so nothing in it would have flagged any of the three.

VI. LIMITATIONS

Observations in §V are simulated from the annotated polygons; §V-C validates the noise model but not the full pipeline on recorded data. We attempted an evaluation on a real building-scale RGB-D dataset with annotated rooms and report a negative result that we believe is useful: because such datasets annotate *rooms inside open-plan buildings* while a 360° depth sensor sees through every doorway, a median 57 % of returns fall outside the polygon being scored, and taking the union of a floor’s rooms still leaves 50 % outside because corridors are unannotated. No granularity makes the annotation and the observation describe the same region, so such a corpus cannot grade a bounded-room method. Finally, the pose-covariance defect of §V-D is measured but not fixed; the cure we argue for—a per-wall offset state with the measured prior—is future work.

VII. CONCLUSION

A layout estimate that a robot can act on needs an uncertainty a robot can check. We have shown that such an uncertainty can be maintained online, that it can be validated against ground truth at scale, and that it is useful precisely because something depends on it. The discipline of making precisions honest is not bookkeeping: it exposed three defects, and it showed that the quantity everyone tunes as a constant is not one.